



AEROSPACE MATERIAL SPECIFICATION

AMS2375™**REV. F**

Issued 1969-05
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Revised 2022-09

Superseding AMS2375E

Control of Forgings Requiring First Article Approval

RATIONALE

AMS2375F is the result of a Five-Year Review and update of the specification. The revision clarifies the approving authority (several locations), adds an example (3.1.1), and addresses grain flow requirements (3.2.3.1).

1. SCOPE

1.1 Purpose

This specification covers procedures for obtaining first-article (preproduction) approval of forgings and the controls to be exercised in producing subsequent production forgings.

1.2 Application

This procedure has been used typically for forgings which require preproduction approval and where assurance is required that production lots are of essentially the same metallurgical quality and are produced by the same basic procedures as the forging originally qualified.

2. APPLICABLE DOCUMENTS

The issue of the following documents in effect on the date of the purchase order forms a part of this specification to the extent specified herein. The supplier may work to a subsequent revision of a document unless a specific document issue is specified. When the referenced document has been cancelled and no superseding document has been specified, the last published issue of that document shall apply.

2.1 SAE Publications

Available from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, Tel: 877-606-7323 (inside USA and Canada) or +1 724-776-4970 (outside USA), www.sae.org.

AS7766 Terms Used in Aerospace Metals Specifications

2.2 ASTM Publications

Available from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959, Tel: 610-832-9585, www.astm.org.

ASTM A370 Mechanical Testing of Steel Products

ASTM E3 Preparation of Metallographic Specimens

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For more information on this standard, visit

<https://www.sae.org/standards/content/AMS2375F/>

- ASTM E8/E8M Tension Testing of Metallic Materials
- ASTM E10 Brinell Hardness of Metallic Materials
- ASTM E18 Rockwell Hardness of Metallic Materials

2.3 Definitions

Terms used in AMS are defined in AS7766.

3. TECHNICAL REQUIREMENTS

3.1 Forging Stock

3.1.1 The forging producer shall determine that the forging stock conforms to the applicable material specification and will yield acceptable forgings. Tests which are characteristic of a heat or lot, e.g., chemical analysis, and that are conducted by the forging stock producer, need not be repeated by the forging producer provided the results of such tests have been supplied by the forging stock producer.

3.1.2 When specified, forging stock shall be procured from sources approved by Cognizant Engineering Organization of the forgings.

3.2 Preproduction Forgings

3.2.1 Procedure

The forging producer shall produce and heat treat to final condition one or more preproduction forgings and shall test a forging or sections thereof to determine conformance to all requirements of the material specification, the drawing, and any additional requirements specified by Cognizant Engineering Organization. The preproduction forgings shall be made by the same practices to be used on production forgings and, unless otherwise specified, may be produced as part of the initial production run. A duplicate preproduction forging or the remaining section of such forgings shall be submitted to Cognizant Engineering Organization for confirmatory testing when requested.

3.2.1.1 When forgings are to be supplied in other than the final heat treated condition, the test forging(s) shall be heat treated to the final heat treated condition of the part. If the as-forged section size is too large to achieve proper heat treatment response, the forging, or a section thereof, shall be machined to a configuration of the same equivalent round as that of the heaviest section of the heat treat configuration.

3.2.1.1.1 If the final heat treated condition of the forgings requires a tensile strength of 200 ksi (1379 MPa) or higher, and the forgings are supplied in a different condition, the producer may excise oversize test coupons from such forgings. Such coupons shall be given the final heat treatment prior to machining to specimen size for testing.

3.2.2 Specimen Locations

The location(s) of coupons that are to be taken for mechanical property tests and the sections for grain flow examination shall be as specified by Cognizant Engineering Organization. The forging producer shall perform the required tests on specimens from these locations. If locations are not specified, the forging producer shall select representative areas for testing and, when required, shall obtain concurrence of Cognizant Engineering Organization. Mechanical property test specimens shall be taken in the transverse direction whenever practicable unless only longitudinal properties are specified in the material specification.

3.2.3 Properties

3.2.3.1 Grain Flow

A representative forging shall be sectioned as indicated on the drawing and the sections suitably etched to show the grain flow. Photographs showing the grain flow pattern shall be made of each section and the photographs shall be identified to indicate the corresponding sections in the forging. The grain flow shall conform to the requirements of the drawing and shall follow the general contour of the forging without reentrant flow lines or end grains. However, flash/parting-line end grains are acceptable. When the specified alloy is not conducive to good grain flow determination, an alternative alloy or melting practice showing good grain flow and having equivalent forgeability shall be substituted and so indicated in the test report.

3.2.3.2 Hardness

Hardness surveys in accordance with ASTM A370 for steels or ASTM E10 or ASTM E18 for all metals shall be made across the heaviest section of test forgings, unless otherwise specified.

3.2.3.3 Microstructure

Sections shall be taken from the fully heat-treated forging at the center of the heaviest usable section and at the surface of the heaviest and thinnest sections and prepared for metallographic examination in accordance with ASTM E3. There shall not be any detrimental microstructural abnormalities related to improper forging or heat treating practice or to forging stock quality. Photographs of the microstructures shall be submitted to Cognizant Engineering Organization and any abnormal microstructural conditions identified.

3.2.3.4 Mechanical Properties

Test specimens from the representative forging(s) shall be tested in accordance with ASTM A370 for steels or ASTM E8/E8M for all metals and shall meet the specified mechanical properties for the part in the final heat treated condition.

3.3 Process Control Factors

Process control factors (see 4.4.2.1), established for producing forgings of each part number, shall be submitted with the results of tests on the preproduction forging(s); when permitted by Cognizant Engineering Organization, the process control factors need not be submitted with the test results, but shall be kept on file for review by Cognizant Engineering Organization.

3.4 Production Forgings

Shall be produced using the same operations, practices, and process control factors used on the approved preproduction forging(s) (see 4.2.2.1.2).

3.4.1 Production forgings shall not be shipped until the forging producer has received written approval of the preproduction forging(s) unless pre-shipment approval is waived by Cognizant Engineering Organization.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection

The producer of forgings shall supply all samples for producer's tests and shall be responsible for the performance of all required tests. Cognizant Engineering Organization reserves the right to sample and to perform any confirmatory testing deemed necessary to ensure that the forgings conform to specified requirements.

4.2 Classification of Tests

4.2.1 Acceptance Tests

Tests of forging stock and forgings to determine conformance to applicable requirements of the material specification (3.2.1) and to any additional requirements specified by Cognizant Engineering Organization are acceptance tests and shall be performed on each heat or lot, as applicable. Such tests of forging stock shall be performed prior to use of a new heat or lot for production forgings.

4.2.2 Preproduction Tests

Tests of preproduction forgings to determine conformance to all applicable technical requirements of the material specification, to requirements for grain flow (3.2.3.1), hardness (3.2.3.2), microstructure (3.2.3.3), and mechanical properties (3.2.3.4), and to any additional requirements specified by Cognizant Engineering Organization are preproduction tests and shall be conducted under the following circumstances:

4.2.2.1 The first production of a new forging.

4.2.2.1.1 At any time a change is made in forging design or in process control factors (4.4.2).

4.2.2.1.2 Forger shall obtain Cognizant Engineering Organization's concurrence on changes that are deemed not to affect preproduction test results including, when applicable, those changes listed in 4.2.2.1.1.

4.3 Sampling and Testing

Shall be as follows; a lot shall be all forgings of the same part number or configuration, opposite hands being considered one configuration, produced from the same heat of forging stock, heated and forged in the same manner, and heat treated in a continuous furnace or in a series of batch-type furnace operations with no change in furnace settings or interruption of power.

4.3.1 For Acceptance Tests

As specified on forging drawing, material specification, or quality assurance specification. For forgings supplied in the final heat treat condition, one forging from each lot shall be subjected to destructive testing. On forgings that contain a test prolongation, each prolongation shall be tested unless a sampling plan is agreed upon by Cognizant Engineering Organization and producer. Each prolongation, and the forging it represents, shall be given a serial number for identification.

4.3.2 For Preproduction Tests

As specified in Section 3, tests shall be performed on the first production of a new forging by a producer and following any change in configuration or forging design.

4.4 Approval

4.4.1 Preproduction forgings and the forging procedure shall be approved by Cognizant Engineering Organization before production forgings are supplied unless preproduction approval is waived by Cognizant Engineering Organization. Neither the approval of preproduction forgings and procedures nor the waiver of preproduction approval shall relieve the forging producer of responsibility for continued conformance to all requirements.

4.4.2 The forging producer shall establish for each part number or configuration parameters for process control factors that will produce forgings conforming to the requirements of this specification and the material specification. These control factors shall constitute the approved manufacturing procedures and shall be used for production forgings. If necessary to make any change in parameters for process control factors, the forging producer shall submit for reapproval a statement of the proposed changes and, when requested, perform preproduction tests of revised forgings. Production forgings incorporating the revised operations shall not be shipped prior to receipt of reapproval unless prior approval is waived by Cognizant Engineering Organization.

4.4.2.1 Process control factors for producing forgings include, but are not limited to, the following:

Type (ingot, bloom, billet, or bar), nominal size (cross-sectional area), shape, and nominal multiple-weight of forging stock; rounds and round-corner-squares of the same nominal cross-sectional area are considered the same shape

Inspection and qualification of forging stock

Processing sequence or number of operations that would result in different cross-sectional structure, grain flow, or working of the metal

Type of forging equipment (press, hammer, ring roll, etc.)

Description of die configuration, dimensions, and forging pressure

Thermal parameters, including heating of stock for forging, annealing, and heat treatment of forgings

Stress relief procedures, when applicable

Protective atmosphere and/or coatings

Cleaning operations (e.g., chemical descaling, blasting)

Inspection procedures

4.4.2.1.1 Any of the above process control factors for which parameters are considered proprietary by the producer may be assigned a code designation. Each variation in such factors shall be assigned a modified code designation. The producer shall maintain complete records of all proprietary processes and factors.

4.5 Records

4.5.1 Maintenance of Facilities

Each forging producer shall keep records demonstrating that the facilities used to produce, control, measure, and test forging stock and forgings during manufacture and inspection are properly maintained and are checked for accuracy at stated intervals against acceptable standards for accuracy. Records shall be maintained on file for not less than 7 years, unless otherwise specified by Cognizant Engineering Organization.

4.5.2 Process Instructions

For each forging part number or configuration, opposite hands being considered a single configuration, the forging producer shall prepare and maintain documented instructions defining the processing methods and routing in manufacturing and inspection cycles.

4.6 Surveillance Visits

The forging producer shall, when requested, permit Cognizant Engineering Organization's authorized personnel to survey all facilities and controls related to production of forgings and to review producer's control of materials and of chemical and metallurgical processing during production.

4.7 Reports

The forging producer shall furnish with the preproduction forging(s) or sections thereof a report showing the results of tests to determine conformance to the technical requirements of this specification, the applicable material specification, and the drawing and shall include photographs of the grain flow and microstructures. The report shall also include the source of forging stock and physical location of the forging producer.

4.8 Resampling and Retesting

If any specimen used in the above tests fails to meet specified requirements, disposition of the forgings shall be based on the resampling and retesting requirements of the applicable material specification. If the material specification does not state the resampling and retesting procedure, disposition of the forgings may be based on the results of testing three additional specimens (two for aluminum and magnesium alloys) for each original nonconforming specimen. Failure of any retest specimen to meet the specified requirements shall be cause for rejection of the forgings represented. Results of all tests shall be reported.

5. PREPARATION FOR DELIVERY

5.1 Identification

Forgings ordered to, and meeting, the requirements of this specification shall be identified with AMS2375F and in accordance with the requirements of the applicable material specification.

5.2 Packaging

Shall be as specified in the applicable material specification.

6. ACKNOWLEDGMENT

A producer shall mention this specification number and its revision letter in all quotations and when acknowledging purchase orders.

7. REJECTIONS

Forgings not conforming to this specification and the applicable material specification, or to modifications authorized by Cognizant Engineering Organization, will be subject to rejection.

8. NOTES

8.1 Revision Indicator

A change bar (I) located in the left margin is for the convenience of the user in locating areas where technical revisions, not editorial changes, have been made to the previous issue of this document. An (R) symbol to the left of the document title indicates a complete revision of the document, including technical revisions. Change bars and (R) are not used in original publications, nor in documents that contain editorial changes only.

8.2 Terms used in AMS are defined in AS7766.

8.3 Dimensions and properties in inch/pound units and the Fahrenheit temperatures are primary; dimensions and properties in SI units and the Celsius temperatures are shown as the approximate equivalents of the primary units and are presented only for information.

8.4 Purchase documents should specify not less than the following:

AMS2375F

Types of test material for confirmatory testing (see 3.2.1)

Rough machined heat treat configuration (see 3.2.1.1)

Tensile test and grain flow locations (see 3.2.2)

PREPARED BY SAE AMS E CARBON AND LOW ALLOY STEELS COMMITTEE